

A Grid of Numbers — Practice AS 9618

Name: _____

Two philosophies and one verb. A bitmap stores *samples*, a vector stores *instructions* — and 9618 asks you to **justify**, which means arguing from the source and the use, never from "better quality".

1. Colour depth. [2]

(a) State what is meant by the **colour depth** of a bitmap image. [1]

(b) An image has a colour depth of **4 bits**. State the number of different colours it can contain. [1] _____

2. Explain the difference between *image resolution* and *screen resolution*. [2]

3. Estimate. A bitmap image has a resolution of **1024 × 768** and a colour depth of **16 bits**. [4]

(a) **Estimate** the file size in mebibytes (MiB). Show all your working. [3]

(b) State why the question asks you to *estimate* rather than to calculate. [1]

4. State three items of data that are stored in the file header of a bitmap image. [3]

1 _____ 2 _____
3 _____

5. Both dials at once. A camera is upgraded. The new sensor **doubles** the width and **doubles** the height of every photograph, and the colour depth changes from **24 bits** to **48 bits**. [2]

State the factor by which the file size increases, and show how you obtained it.

6. A file, decoded. A bitmap image has a resolution of 4 × 4 and a colour depth of 2 bits. Each byte below is one row, leftmost pixel first. [4]

14 7D 7D 14

(a) Complete the grid using the palette. [2] (b) State the file size in bytes. [1] (c) State how many bytes this image would need at a colour depth of 24 bits. [1]

14				
7D				
7D				
14				

value	colour
00	white
01	light
10	dark
11	black

(b) _____
(c) _____
Each byte is 8 bits and each pixel takes 2 of them. Count carefully.

7. Vector graphics. Describe each of the following terms. [3]
drawing object

property

drawing list

8. Justify. [4]

A company is preparing two files. **File A** is a photograph of the head office, for the "about us" page of its website. **File B** is the company logo, which must appear on a business card and on a banner across the front of the building.

For **each** file, state whether a bitmap image or a vector graphic should be used, and **justify** your choice.
File A

File B

 **No ice-cream challenge this lesson, and that is deliberate.** The code is in the deck and it is worth re-running: the 8 × 8 bitmap you edit by hand, the colour that comes out as 0x2563eb, and the PNG header where P N G falls out of four raw numbers. Change a value in each and watch what breaks.

— **Want to dig?** Question 3 told you to say *estimate* because of the header — but nobody ever tells you how big a header actually is. Go and find out for one real format (a BMP header is a good place to start, and it is a fixed size). Then answer the question the syllabus never asks: **at what image resolution does the header stop mattering?** Work out the header as a percentage of the file for a 16×16 icon, and then for a 4000×3000 photograph. One of those two answers is the reason the exam ignores it, and the other is the reason the word *estimate* is there at all.

Additional space. If you need more room for any answer, continue here and write the question number in the margin beside it.